

Ce projet consiste à déployer une architecture réseau virtualisée complète et sécurisée à partir de Proxmox, sur laquelle seront installés :

Un routeur/pare-feu pfSense, au centre de l'architecture

Un réseau LAN pour les postes clients

Une DMZ contenant un serveur Web exposé

Un VLAN de management strictement réservé à l'administration

Un système de NAT, contrôles d'accès, proxy filtrant, et début de supervision

Le but est non seulement de déployer ces environnements, mais aussi de mettre en œuvre des mesures de sécurité avancées, telles que :

Isolation stricte des zones LAN ↔ DMZ

Filtrage L3/L4

NAT et redirections de ports contrôlées

Administration restreinte (VLAN 99)

Proxy filtrant (Squid + SquidGuard)

Préparation à la supervision centralisée

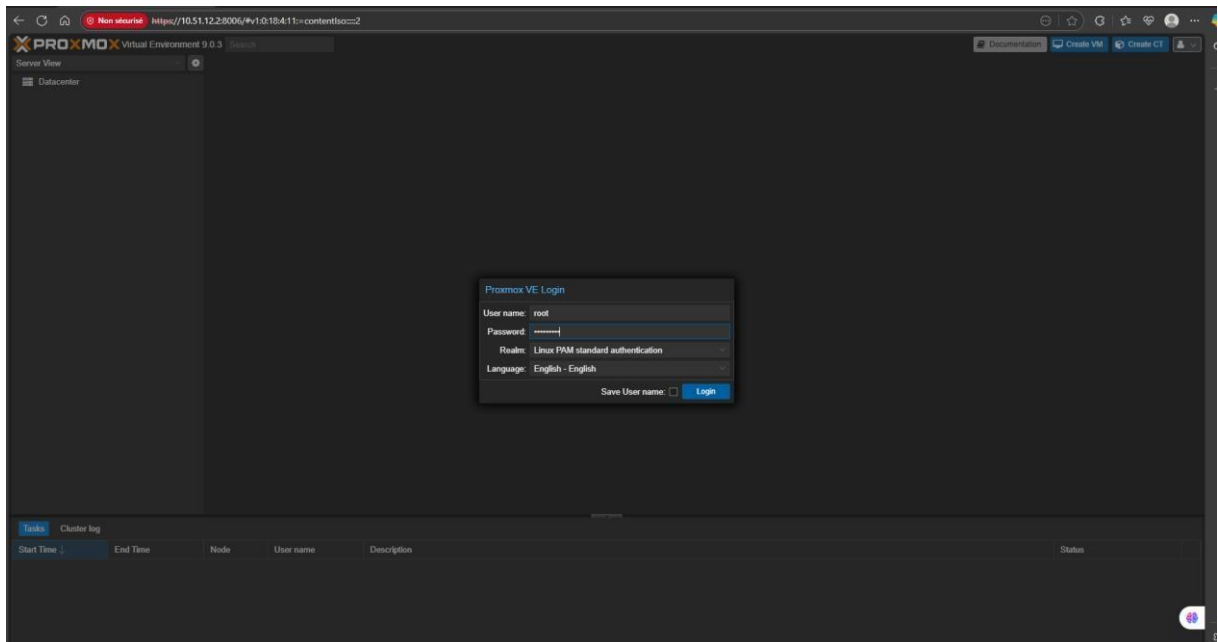
1. Installation de Proxmox

Installation de Proxmox sur serveur/PC dédié.

Partitionnement automatique.

Création du mot de passe root + email.

Choix du réseau (interface principale = future WAN).



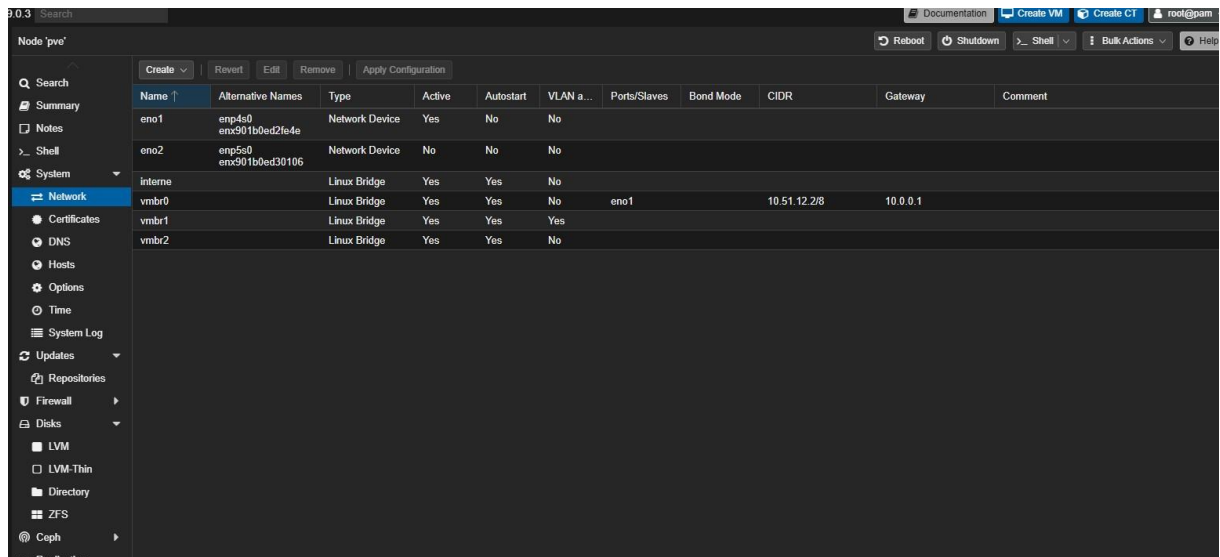
2. Création des bridges réseau

Objectif : isoler les réseaux.

Bridge Rôle

vmbr0 WAN (accès Internet / réseau physique) 10.50.0.53

vmbr1 LAN (postes clients) 192.168.1.1 vmbr2 DMZ
(serveurs exposés) 192.168.2.1



Premier accès à l'interface Web

Depuis un navigateur g mis : <https://10.50.0.53:8006>

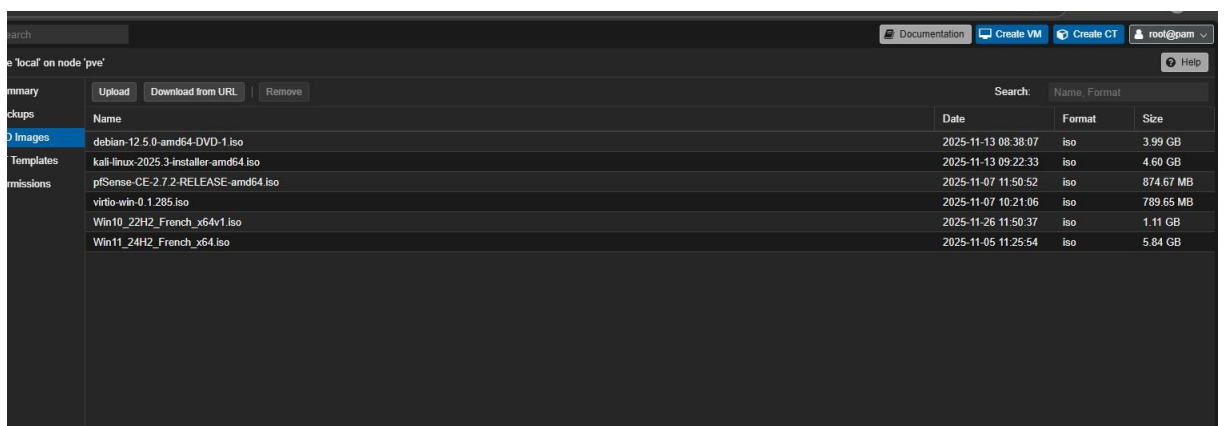
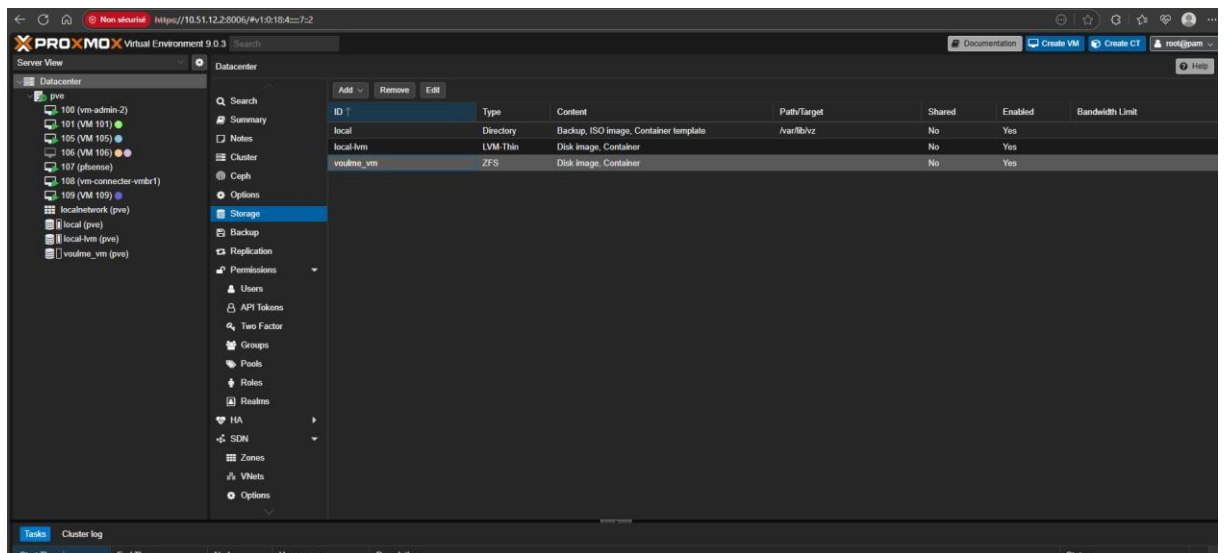
L'interface permet :

- Gestion des VMs
- Gestion des réseaux
- Gestion du stockage
- Snapshots
- Console Web des VMs

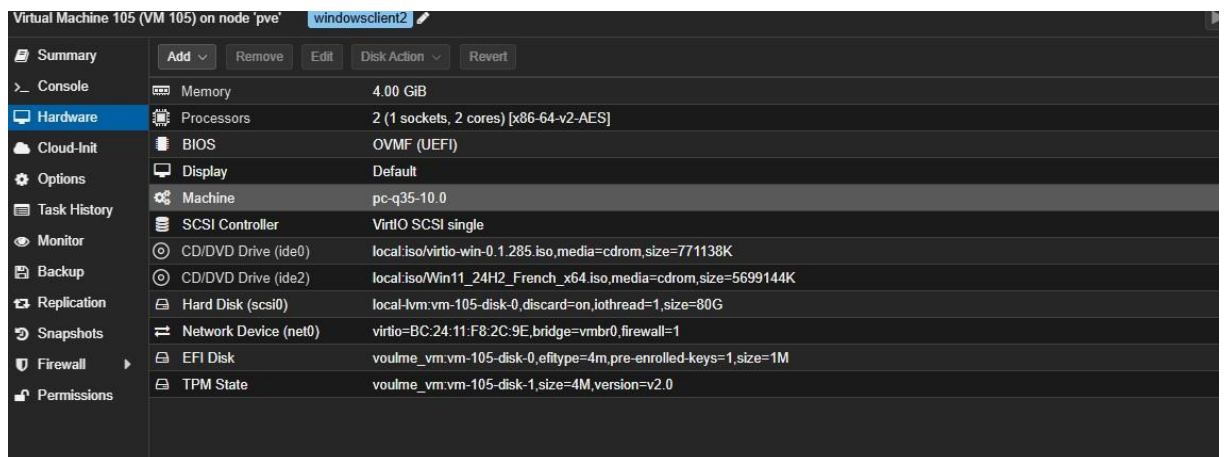
4. Vérification du stockage

J'ai créé un espace de stockage `volum_vm` pour stocker toutes les VM et je leur ai attribué les tailles suffisantes :

- allocation dynamique
- snapshots rapides
- optimisation de l'espace



5. Creation de la vm



II. Mise en place du routeur pfSense

1. Création de la VM pfSense

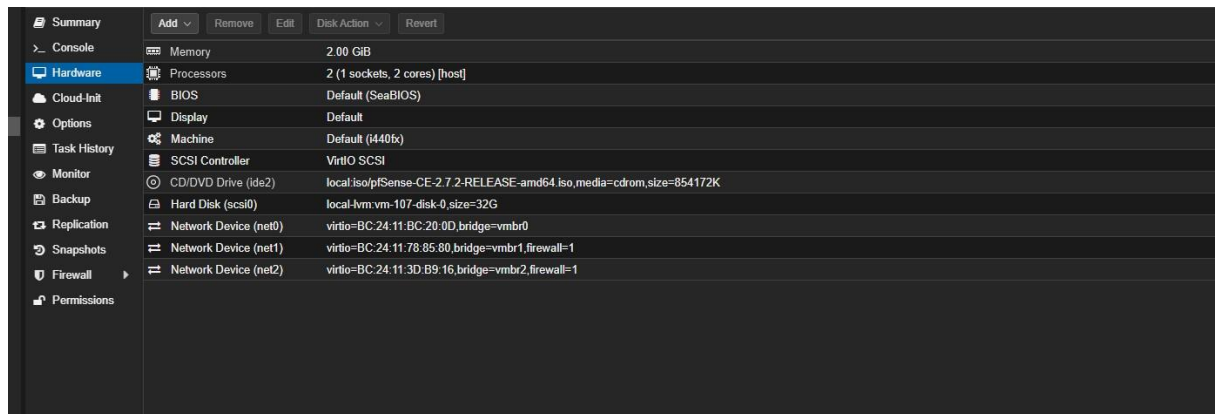
- 2 vCPU
- 2Go RAM
- 32 Go disque

- 3 cartes réseau :
 - net0 → vmbr0 (WAN) ◦

net1 → vmbr1 (LAN) ◦ net2 →

vmbr2 (DMZ) Formats :

- Sea bios
- VirtIO pour le disque et les interfaces réseau



- J'ai ensuite assigner les interfaces WAN → vtnet0
- LAN → vtnet1
- OPT1 → vtnet2

```
pfSense 2.7.2-RELEASE amd64 20231206-2010
Bootup complete

FreeBSD/amd64 (pfSense12.proxmox12.arpa) (ttyv0)

QEMU Guest - Netgate Device ID: cd1cf2aa456c37eb659d

*** Welcome to pfSense 2.7.2-RELEASE (amd64) on pfSense12 ***

WAN (wan)      -> vtnet0      -> v4/DHCP4: 10.50.0.53/8
LAN (lan)      -> vtnet1      -> v4: 192.168.1.1/24
DMZ (opt1)     -> vtnet2      -> v4: 192.168.2.1/24
MANAGEMENT (opt2) -> vtnet1.99 -> v4: 192.168.99.1/24

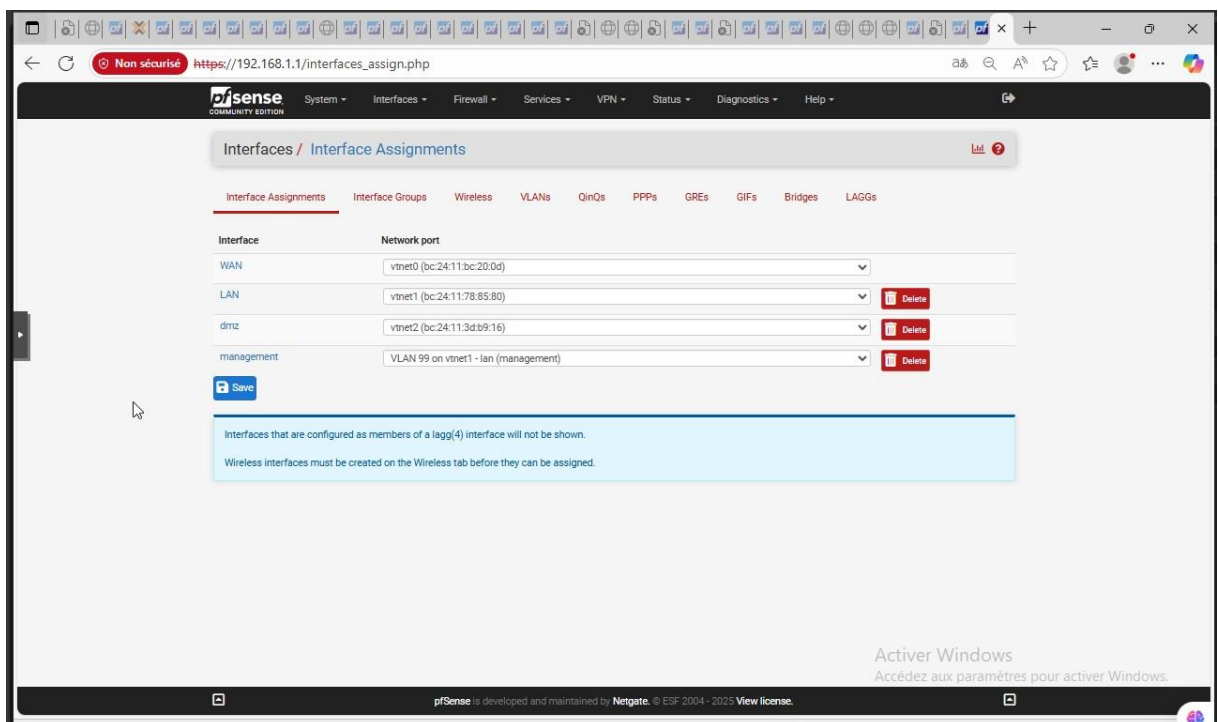
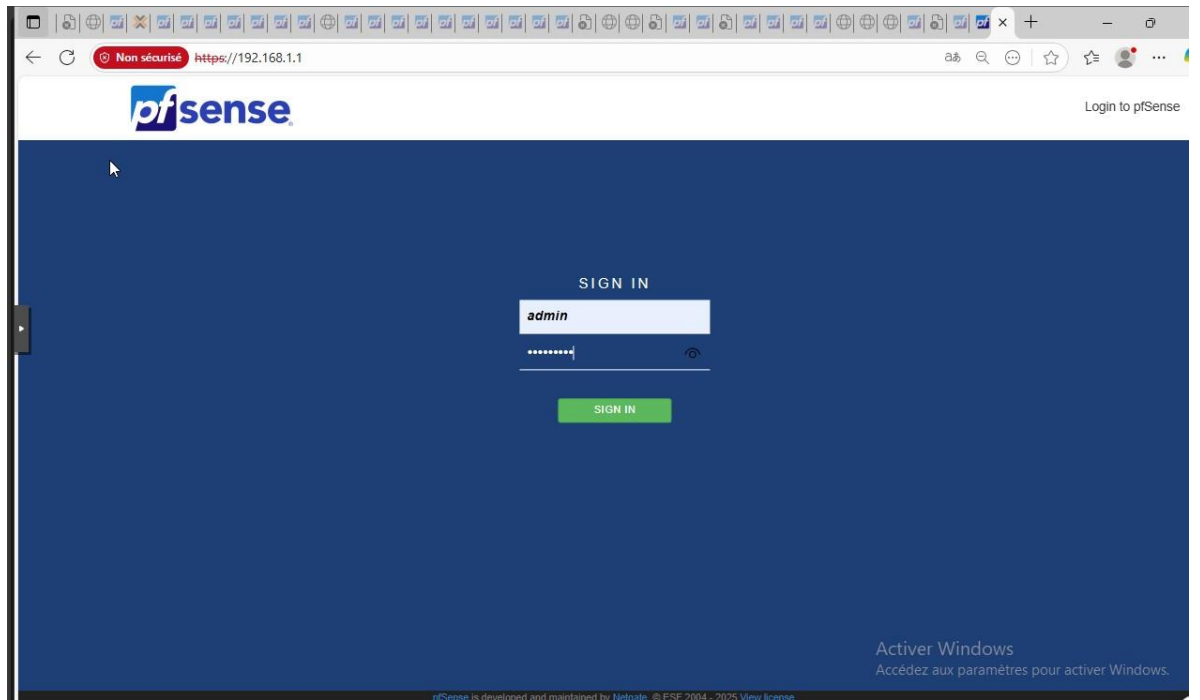
0) Logout (SSH only)          9) pfTop
1) Assign Interfaces          10) Filter Logs
2) Set interface(s) IP address 11) Restart webConfigurator
3) Reset webConfigurator password 12) PHP shell + pfSense tools
4) Reset to factory defaults  13) Update from console
5) Reboot system              14) Enable Secure Shell (sshd)
6) Halt system                15) Restore recent configuration
7) Ping host                  16) Restart PHP-FPM
8) Shell
```

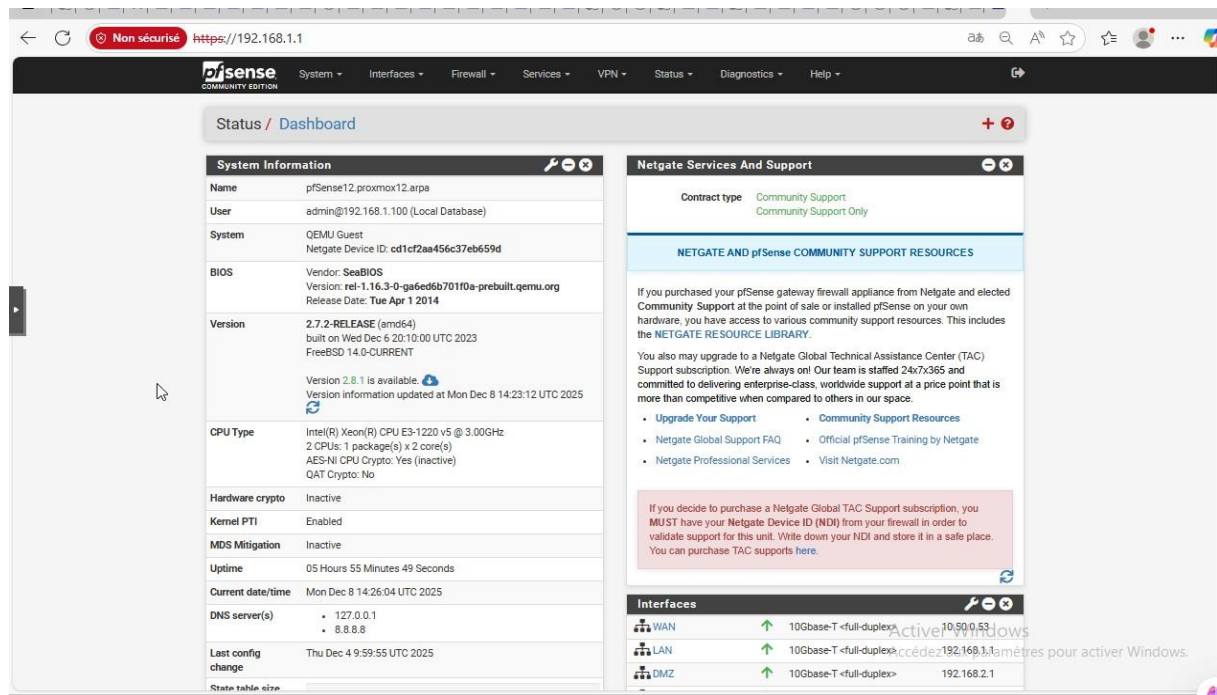
4. Accès Web à pfSense

Avec ma vm connecter a vmbr1 à qui j'ai attribuer seulement la carte reseau vmbr1 (lan)

Summary	Add	Remove	Edit	Disk Action	Revert
> Console	Memory	4.00 GiB			
Hardware	Processors	2 (1 sockets, 2 cores) [x86-64-v2-AES]			
Cloud-Init	BIOS	OVMF (UEFI)			
Options	Display	Default			
Task History	Machine	pc-q35-10.0			
Monitor	SCSI Controller	VirtIO SCSI single			
Backup	CD/DVD Drive (ide0)	local:iso/virtio-win-0.1.285.iso,media=cdrom,size=771138K			
Replication	CD/DVD Drive (ide2)	local:iso/Win11_24H2_French_x64.iso,media=cdrom,size=5699144K			
Snapshots	Hard Disk (scsi0)	local-lvm:vm-108-disk-0,discard=on,iothread=1,size=80G			
Firewall	Network Device (net1)	virtio=BC:24:11:F4:8C:FD,bridge=vmbr1,firewall=1			
Permissions	EFI Disk	voulme_vm:vm-108-disk-0,efitype=4m,pre-enrolled-keys=1,size=1M			
	TPM State	voulme_vm:vm-108-disk-1,size=4M,version=v2.0			

Je suis aller sur pfsense en rentrant l'ip <https://192.168.1.1>:8006





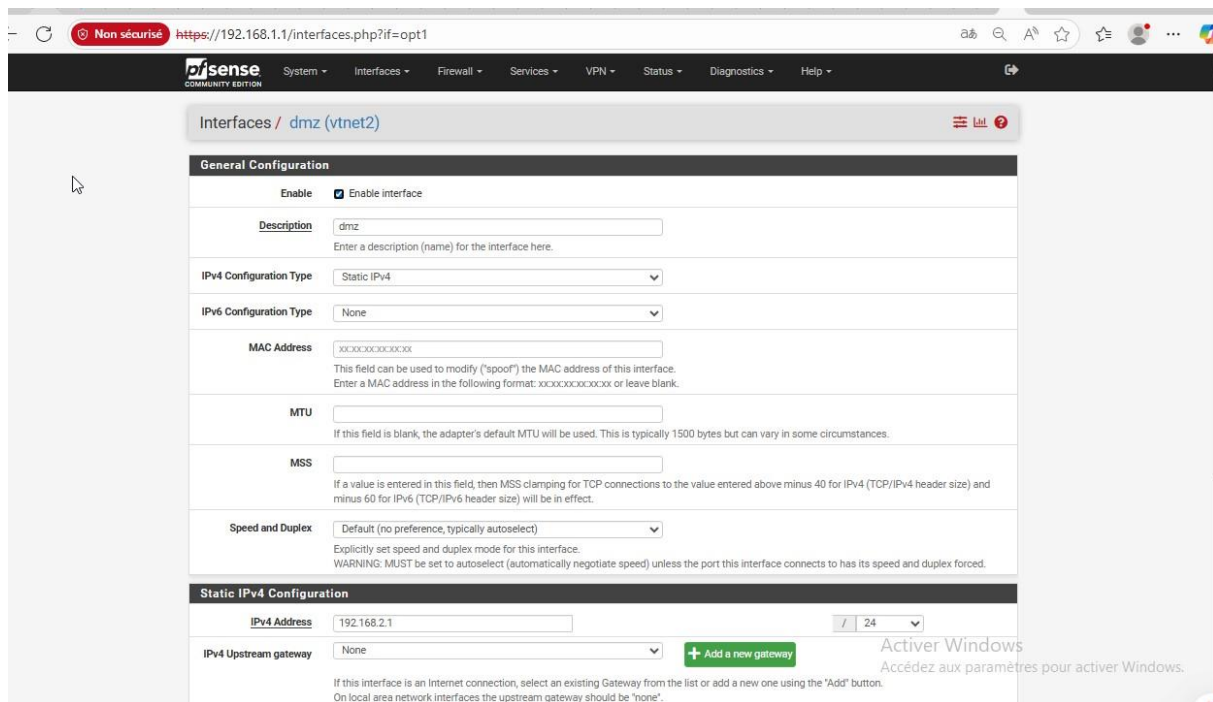
III. Mise en place de la DMZ + Serveur Web

La DMZ doit être isolée du LAN, accessible depuis WAN via NAT contrôlé, et héberger un serveur web.

1. Activation de l'interface OPT1 → DMZ

pfSense → Interfaces → OPT1 Paramètres :

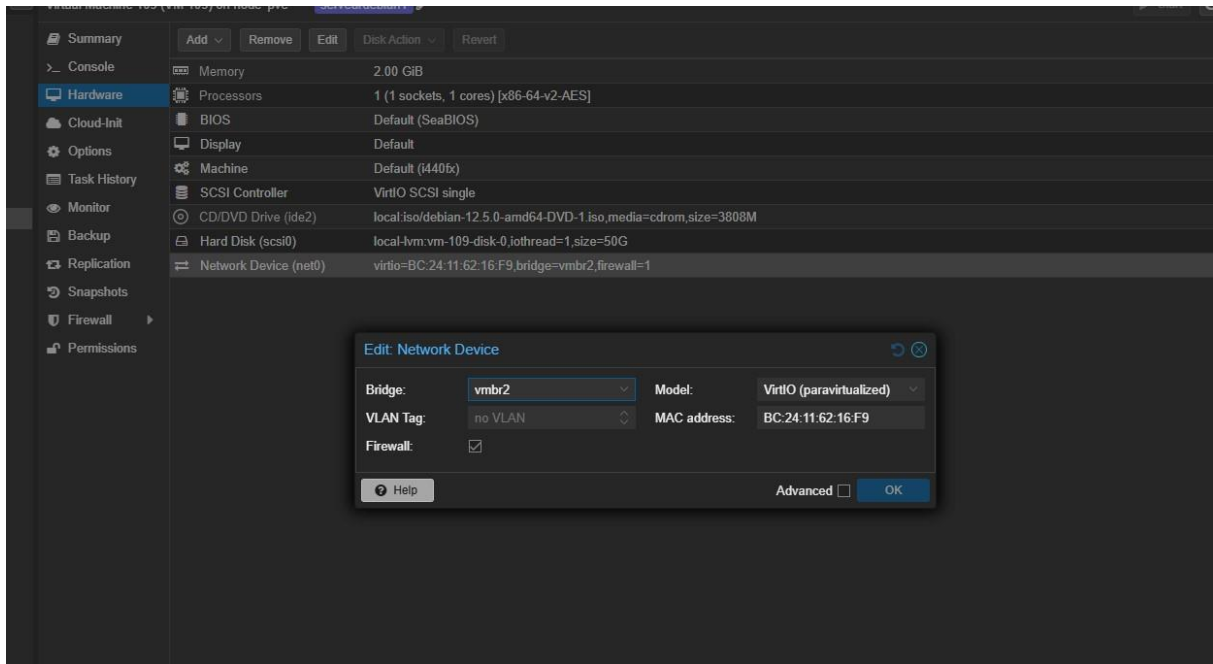
- Enable Interface
- Description : DMZ
- Static IPv4 : 192.168.2.1/24
- DHCP : désactivé



2. Création d'une VM Debian pour la DMZ Interface → vmbr2.

Configuration réseau statique :

```
youssef@debian:~$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: ens18: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether bc:24:11:62:16:f9 brd ff:ff:ff:ff:ff:ff
    altname enp0s18
    inet 192.168.2.10/24 brd 192.168.2.255 scope global ens18
        valid_lft forever preferred_lft forever
    inet6 fe80::be24:11ff:fe62:16f9/64 scope link
        valid_lft forever preferred_lft forever
youssef@debian:~$
```



Ensuite , j'ai installé un serveur web avec nginx

```
youssef@debian:~$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: ens18: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether bc:24:11:62:16:f9 brd ff:ff:ff:ff:ff:ff
    altname enp0s18
    inet 192.168.2.10/24 brd 192.168.2.255 scope global ens18
        valid_lft forever preferred_lft forever
    inet6 fe80::be24:11ff:fe62:16f9/64 scope link
        valid_lft forever preferred_lft forever
youssef@debian:~$
```

```
youssef@debian:~$ curl http://localhost
<!DOCTYPE html>
<html>
<head>
<title>Welcome to nginx!</title>
<style>
html { color-scheme: light dark; }
body { width: 35em; margin: 0 auto;
font-family: Tahoma, Verdana, Arial, sans-serif; }
</style>
</head>
<body>
<h1>Welcome to nginx!</h1>
<p>If you see this page, the nginx web server is successfully installed and
working. Further configuration is required.</p>

<p>For online documentation and support please refer to
<a href="http://nginx.org/">nginx.org</a>.<br/>
Commercial support is available at
<a href="http://nginx.com/">nginx.com</a>.</p>

<p><em>Thank you for using nginx.</em></p>
</body>
</html>
youssef@debian:~$
```

4. Mise en place du NAT : WAN → DMZ

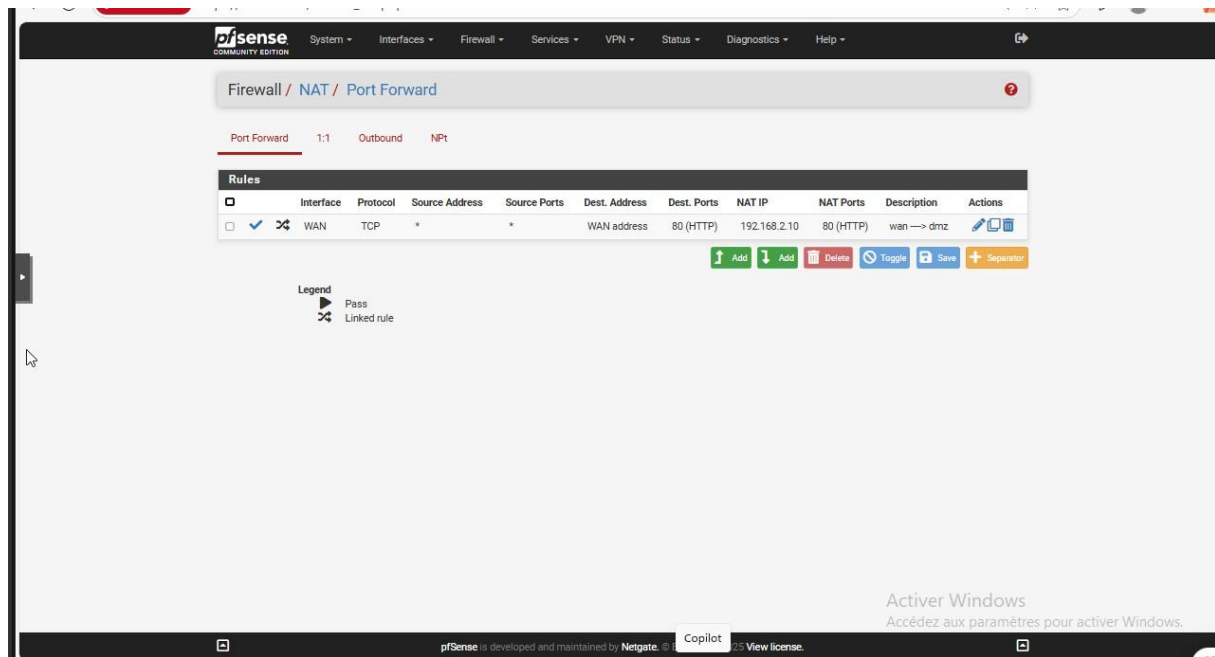
Objectif : rediriger les requêtes du WAN vers 192.168.2.10.

Configuration :

pfSense → Firewall → NAT → Port Forward :

Paramètre	Valeur
Interface	WAN
Protocol	TCP
Destination	WAN address
Dest. Port	80
Redirect target	192.168.2.10
Redirect Port	80

Une règle firewall WAN correspondante est automatiquement créée.



5. Sécurisation : isolation LAN / DMZ

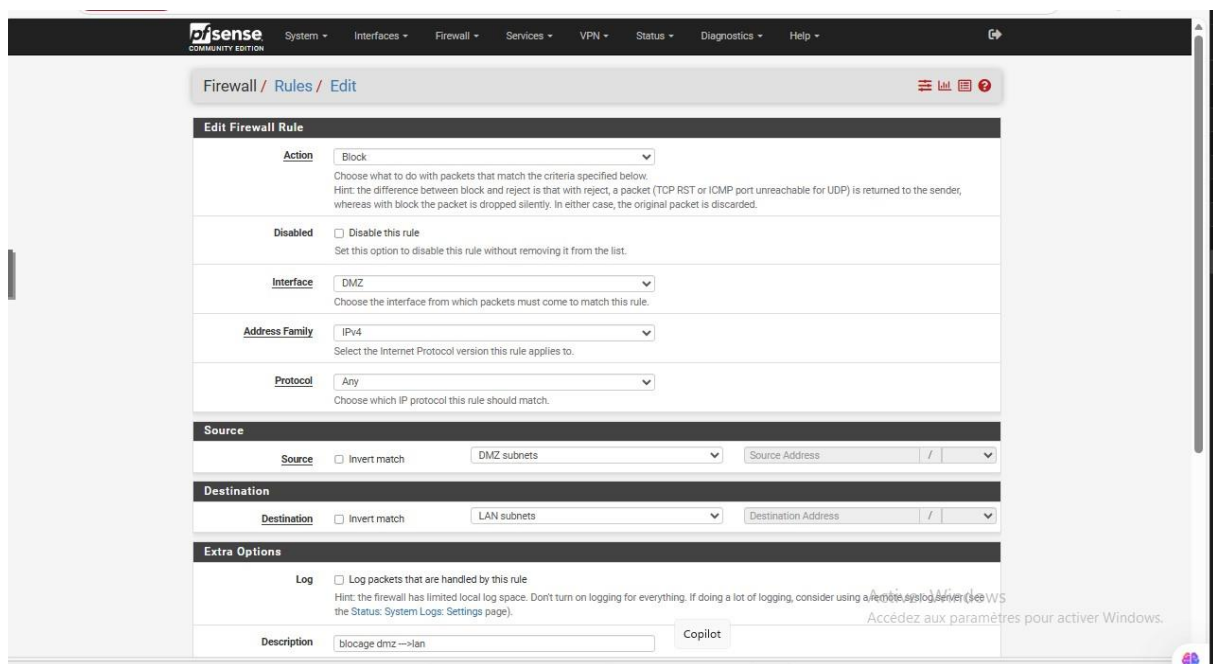
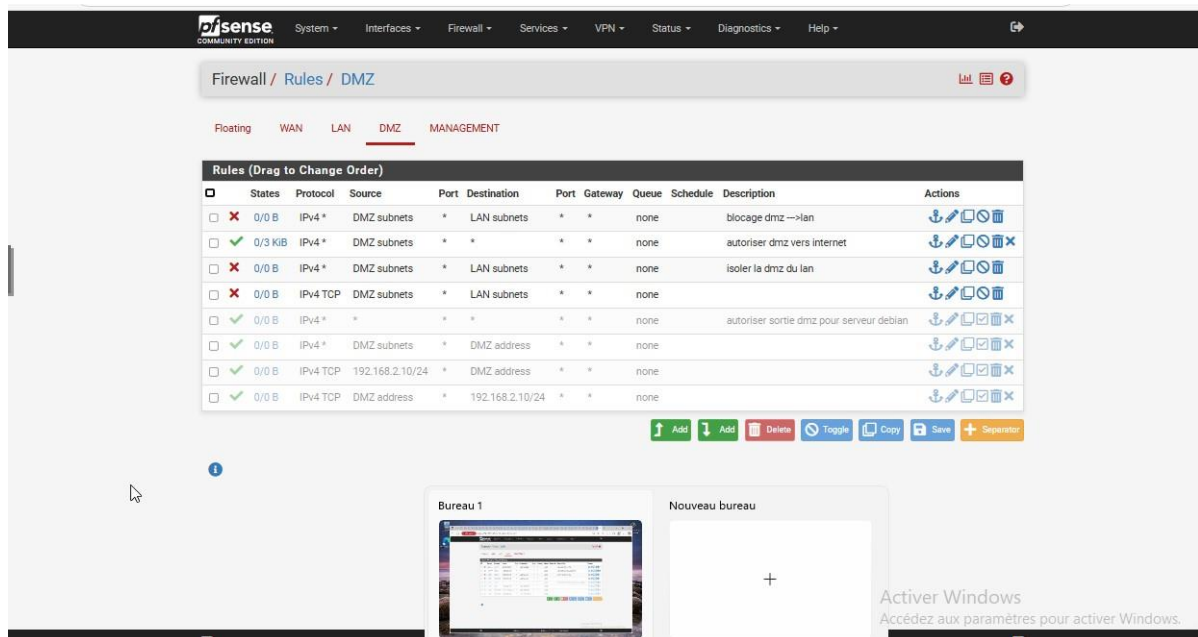
5.1. G Bloquer DMZ → LAN

pfSense → Firewall → Rules → DMZ :

Source : DMZ subnet

Destination : LAN subnet

Action : Block

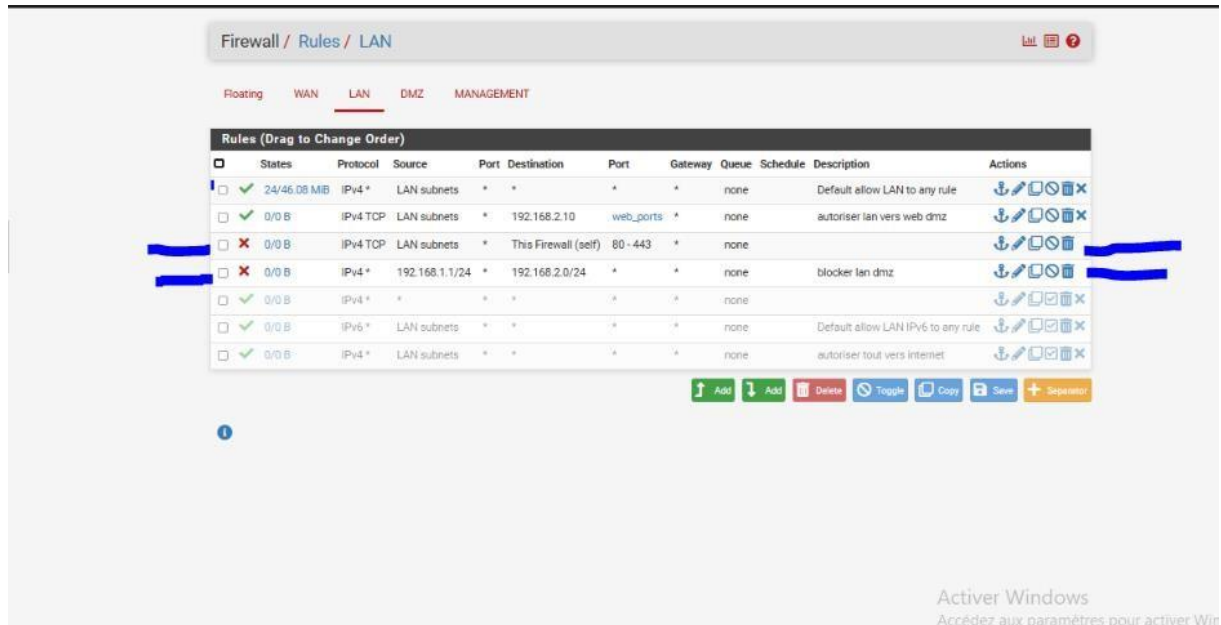


5.2. Limiter LAN → DMZ

Par défaut, LAN peut tout faire → il faut restreindre.

✓ Autoriser LAN → DMZ (HTTP/HTTPS)

✗ Bloquer tout le reste



(J'ai modifier le bon ordre est):

1 **BLOCK** LAN net → DMZ net

2 **ALLOW** LAN → Web DMZ

3 **ALLOW** LAN → Internet 4

ALLOW LAN → *

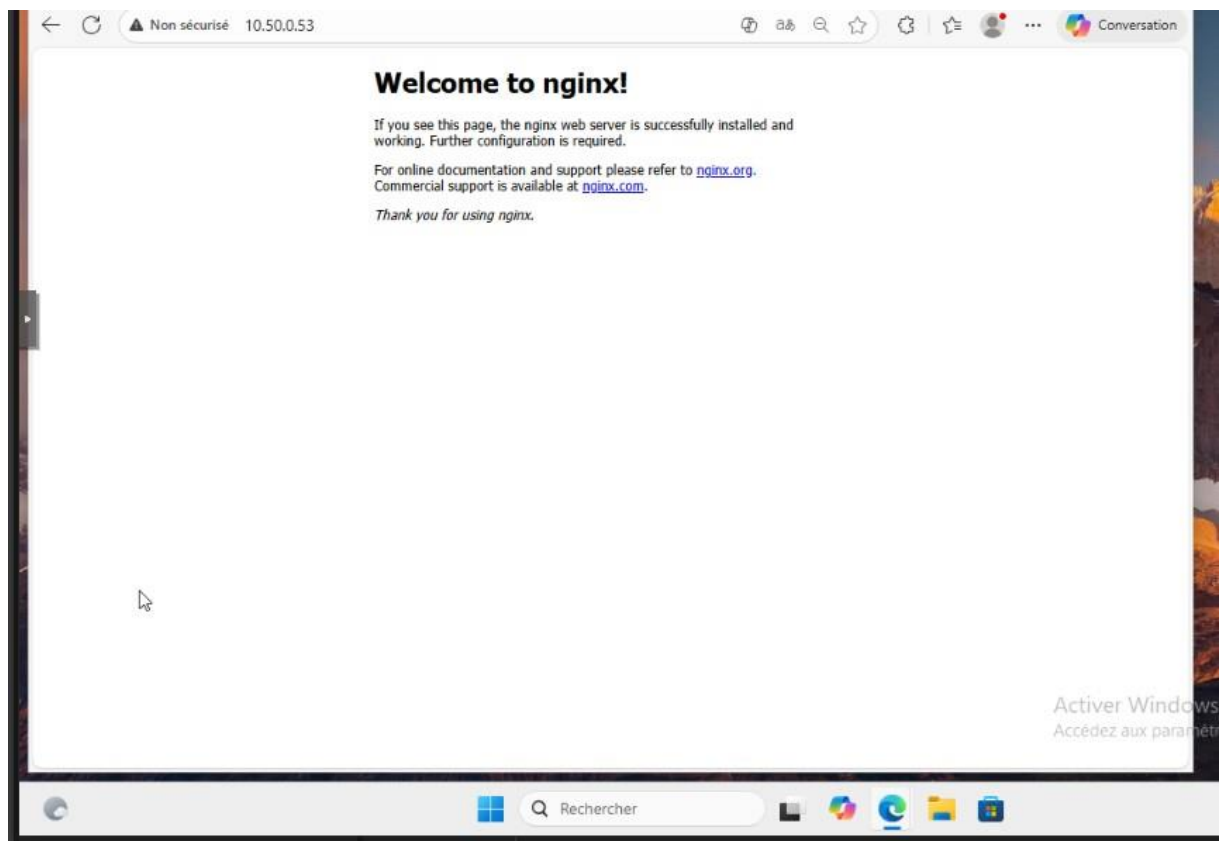
5.3. Vérifications

Depuis DMZ → ping LAN : **bloqué**

Depuis WAN → DMZ (HTTP) : **OK**

```
youssef@debian:~$ ping 192.168.1.1
PING 192.168.1.1 (192.168.1.1) 56(84) bytes of data.

^C
--- 192.168.1.1 ping statistics ---
24 packets transmitted, 0 received, 100% packet loss, time 23535ms
youssef@debian:~$
```



Un pc connecter au wan tombe bien sur la page nginx du serveur debian grace à la regle de par feu de redirection

IV. Sécurisation avancée

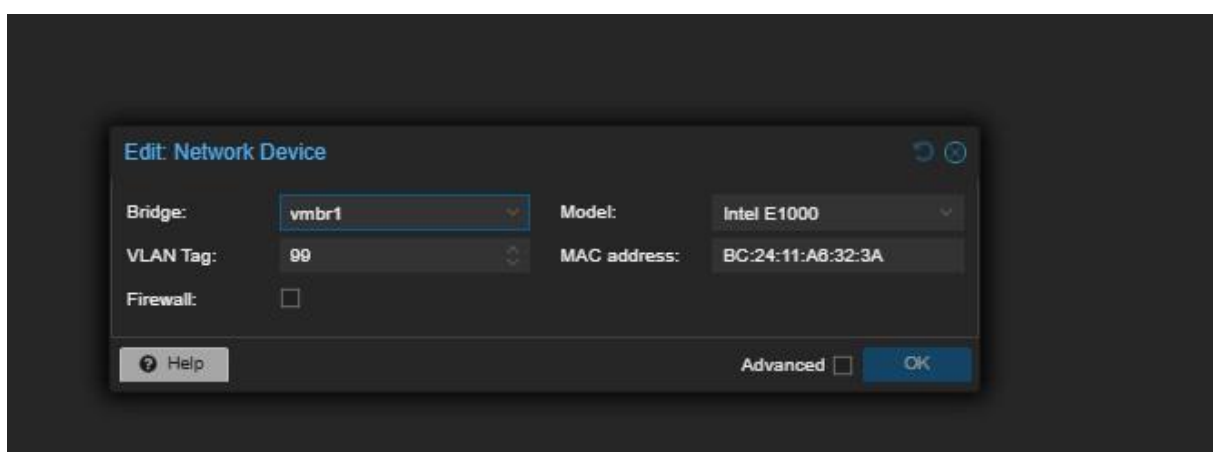
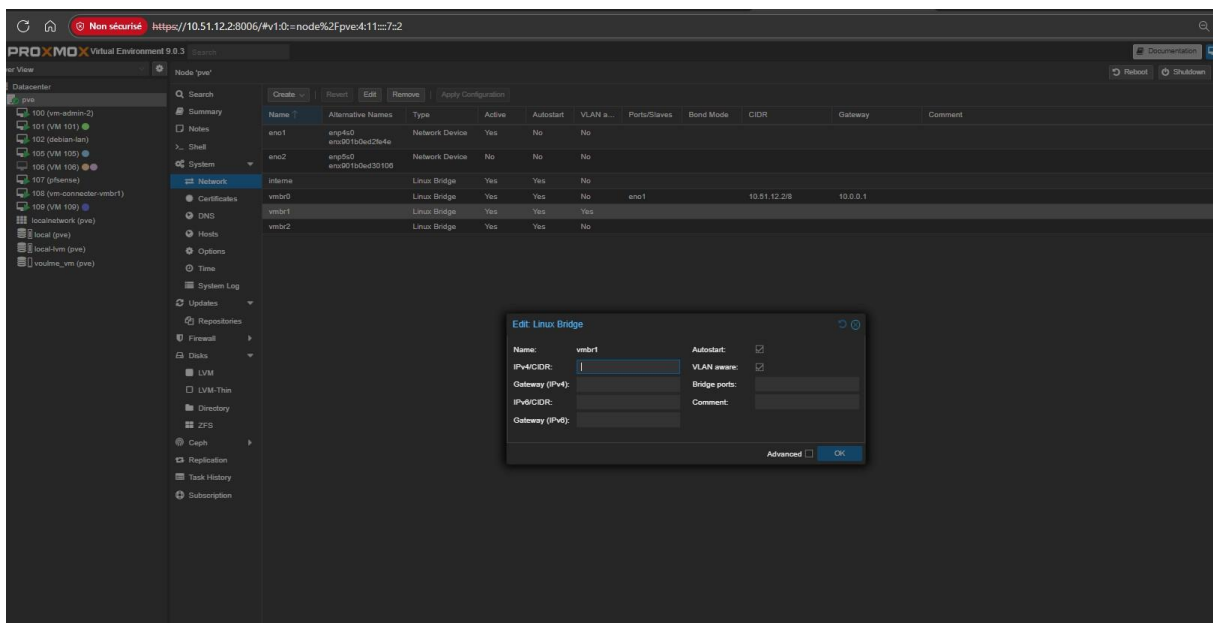
1. Mise en place du VLAN 99 – Management

But :

→ Seule une machine dédiée (VM-admin) peut administrer pfSense.

1.1. Création du VLAN dans Proxmox Proxmox → Datacenter → Network :

- Ajouter VLAN 99 sur vmbr1
- Créer une VM avec interface taguée VLAN 99



1.2. Configuration de l'interface MANAGEMENT pfSense → Interfaces → Assign → Add VLAN :

- Parent : vtnet1
- VLAN tag = 99

Nouvelle interface → MANAGEMENT :

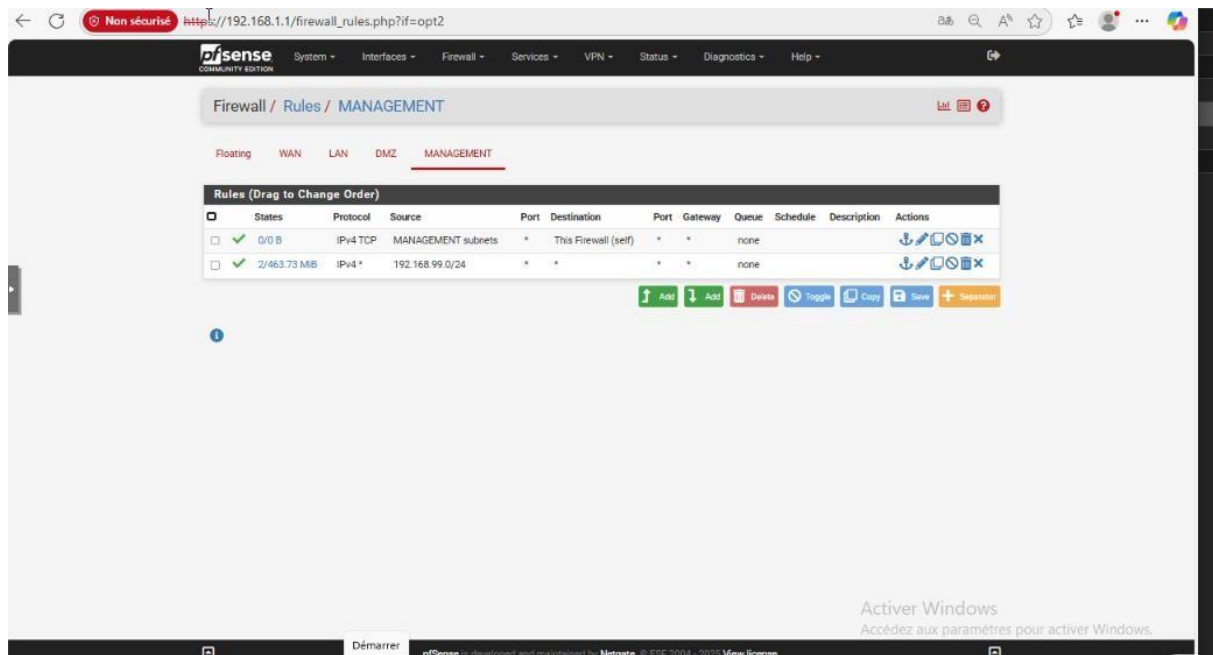
- IP = **192.168.99.1/24**
- DHCP désactivé (VM admin configurée manuellement)

The screenshot displays the pfSense web interface for configuring the 'management' interface (vtnet1.99). The 'General Configuration' section includes fields for 'Enable' (checked), 'Description' (management), 'IPv4 Configuration Type' (Static IPv4), 'IPv6 Configuration Type' (None), 'MAC Address' (00:00:00:00:00:00), 'MTU' (1500), 'MSS' (60), and 'Speed and Duplex' (Default). The 'Static IPv4 Configuration' section shows the 'IPv4 Address' set to 192.168.99.1 and the 'IPv4 Upstream gateway' set to None. A green button labeled 'Add a new gateway' is visible next to the gateway dropdown.

1.3. Règles du VLAN Management

Objectif :

Autoriser uniquement l'accès à pfSense (port 443) depuis 192.168.99.0/24. pfSense → Firewall → Rules → MANAGEMENT :



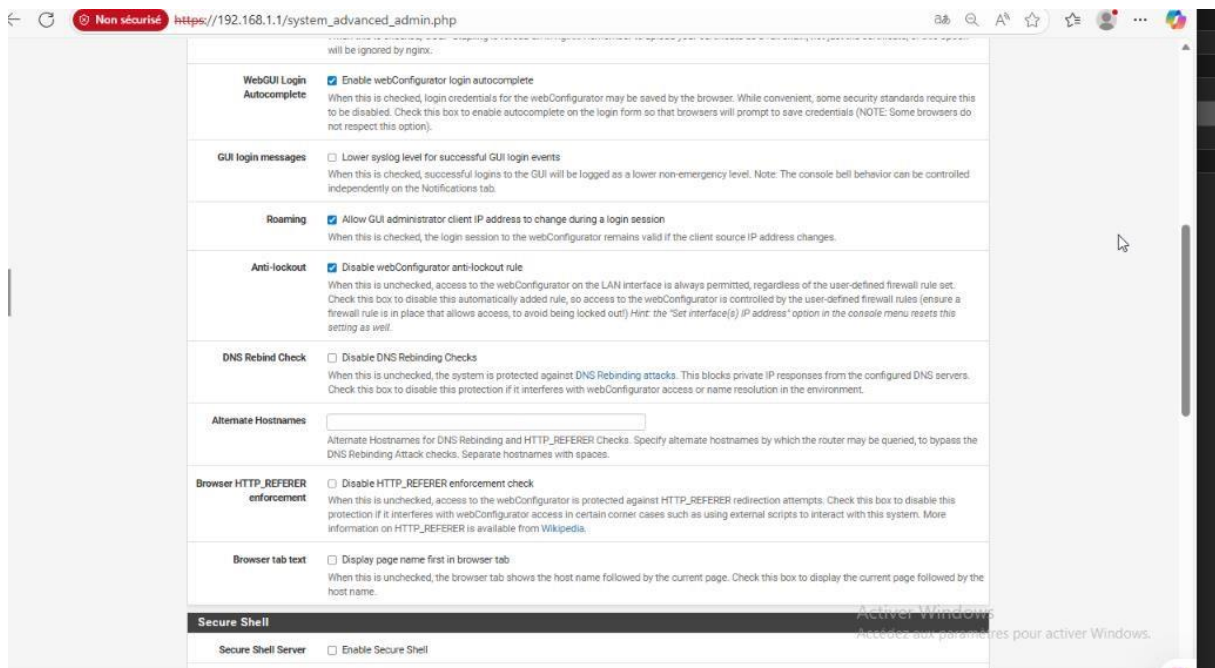
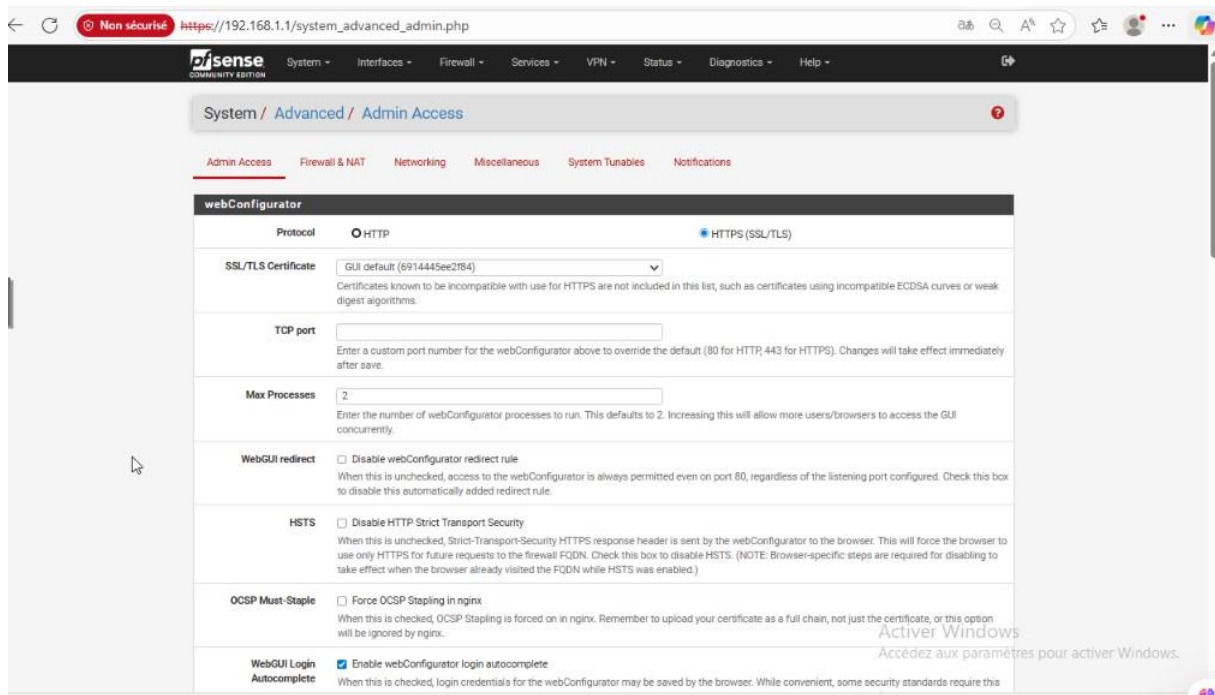
2. Durcissement de l'interface Web pfSense pfSense -

⑦ System → Advanced → Admin Access :

- Activer HTTPS uniquement
- Désactiver HTTP
- Désactiver SSH sur toutes les autres interfaces
- Interface d'écoute : Management seulement

Conséquence :

- Impossible d'administrer pfSense depuis LAN ou DMZ.
- Seule la VM admin VLAN 99 peut y accéder.



V. Proxy Squid + Filtrage SquidGuard

1. Pourquoi mettre un proxy filtrant ? Le proxy permet :

- De contrôler les accès Web sortants
- D'appliquer des politiques : blocage réseaux sociaux, streaming, catégories...
- De journaliser finement les requêtes
- De réduire la charge réseau via du cache
- De surveiller la navigation élèves / utilisateurs

Squid = proxy HTTP(s)

SquidGuard = module de filtrage catégorisé

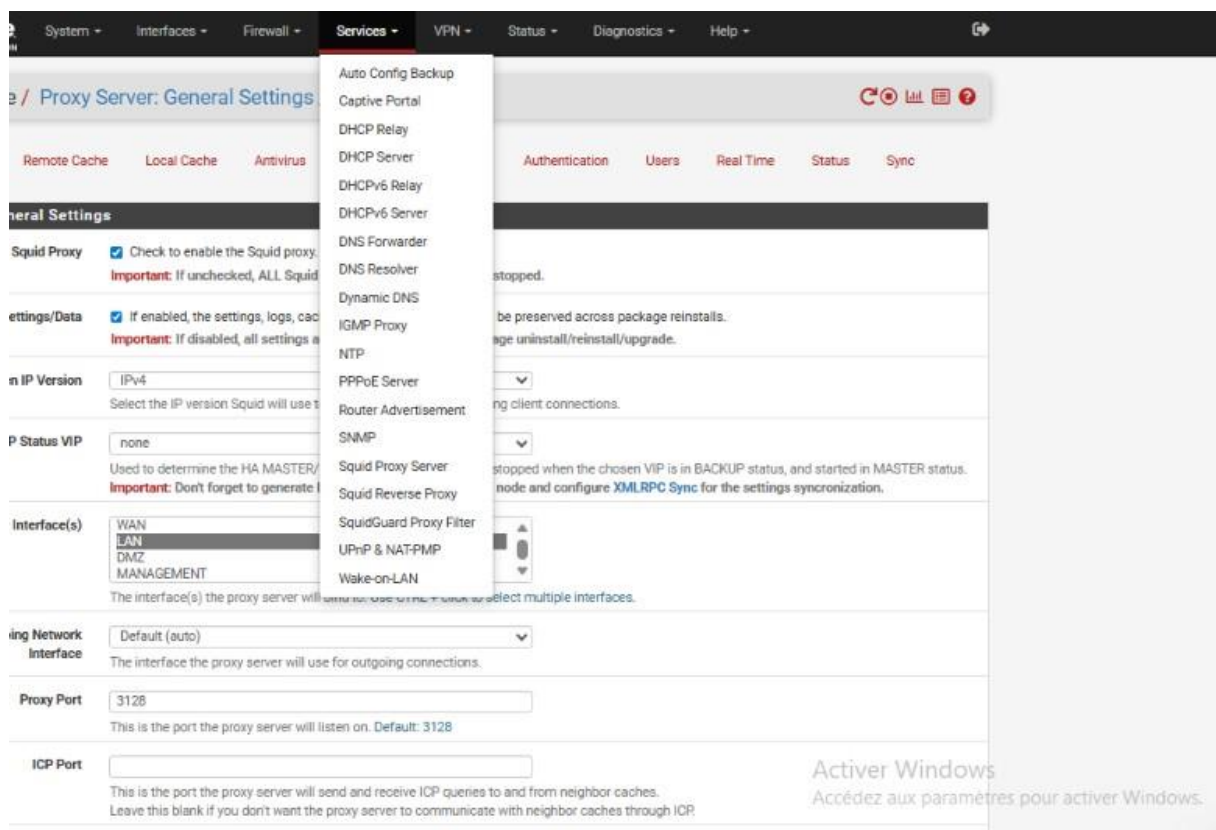
2. Installation du paquet Squid

Menu : **System** → **Package Manager** → **Available Packages**

Installer :

✓ squid

✓ squidGuard



3. Configuration du proxy

Menu : Services → Squid Proxy Server

3.1. Onglet “General”

- Enable Squid
- Proxy mode : **Transparent HTTP / or explicit**
- Ports : 3128 (défaut)
- Allow users on LAN : Oui

Non sécurisé https://192.168.1.1/pkg_edit.php?xml=squid.xml&id=0

General Remote Cache Local Cache Antivirus ACLs Traffic Mgmt Authentication Users Real Time Status Sync

Squid General Settings

Enable Squid Proxy ☒ Check to enable the Squid proxy.
Important: If unchecked, ALL Squid services will be disabled and stopped.

Keep Settings/Data ☒ If enabled, the settings, logs, cache, AV defs and other data will be preserved across package reinstalls.
Important: If disabled, all settings and data will be wiped on package uninstall/reinstall/upgrade.

Listen IP Version IPv4
Select the IP version Squid will use to select addresses for accepting client connections.

CARP Status VIP none
Used to determine the HA MASTER/BACKUP status. Squid will be stopped when the chosen VIP is in BACKUP status, and started in MASTER status.
Important: Don't forget to generate Local Cache on the secondary node and configure XMLRPC Sync for the settings synchronization.

Proxy Interface(s) WAN LAN DMZ MANAGEMENT
The interface(s) the proxy server will bind to. Use CTRL + click to select multiple interfaces.

Outgoing Network Interface Default (auto)
The interface the proxy server will use for outgoing connections.

Proxy Port 3128
This is the port the proxy server will listen on. Default: 3128.

ICP Port
This is the port the proxy server will send and receive ICP queries to and from neighbor caches. Leave this blank if you don't want the proxy server to communicate with neighbor caches through ICP.

Allow Users on Interface ☒ If checked, the users connected to the interface(s) selected in the 'Proxy interface(s)' field will be allowed to use the proxy. There will be no need to add the interface's subnet to the list of allowed subnets.

Patch Captive Portal This feature was removed - see Bug #5594 for details!

Resolve DNS IPv4 First ☐ Enable this to force DNS IPv4 lookup first.

Patch Captive Portal This feature was removed - see Bug #5594 for details!

Resolve DNS IPv4 First ☐ Enable this to force DNS IPv4 lookup first.
This option is very useful if you have problems accessing HTTPS sites.

Disable ICMP ☐ Check this to disable Squid ICMP ping helper.

Use Alternate DNS Servers for the Proxy Server
To use DNS servers other than those configured in System > General Setup, enter the IP(s) here. Separate entries by semi-colons (;)

Extra Trusted CA none
Select extra Trusted CA certificate in addition to the default root certificate bundle.
Warning: This option may only be required if the upstream proxy is using SSL/MITM mode and could be a security issue in other cases.

Transparent Proxy Settings

Transparent HTTP Proxy ☒ Enable transparent mode to forward all requests for destination port 80 to the proxy server.
Transparent proxy mode works without any additional configuration being necessary on clients.
Important: Transparent mode will filter SSL (port 443) if you enable 'HTTPS/SSL Interception' below.
Hint: In order to proxy both HTTP and HTTPS protocols without intercepting SSL connections, configure WPAD/PAC options on your DNS/DHCP servers.

Transparent Proxy Interface(s) WAN LAN DMZ MANAGEMENT
The interface(s) the proxy server will transparently intercept requests on. Use CTRL + click to select multiple interfaces.

Bypass Proxy for Private Address Destination ☐ Do not forward traffic to Private Address Space (RFC 1918 and IPv6 ULA) destinations.
Destinations in Private Address Space (RFC 1918 and IPv6 ULA) are passed directly through the firewall, not through the proxy server.

Bypass Proxy for These Source IPs
Do not forward traffic from these source IPs, CIDR nets, hostnames, or aliases through the proxy server but let it pass directly through the firewall.
Applies only to transparent mode. Separate entries by semi-colons (;)

Bypass Proxy for These Destination IPs
Do not proxy traffic going to these destination IPs, CIDR nets, hostnames, or aliases, but let it pass directly through the firewall.
Applies only to transparent mode. Separate entries by semi-colons (;)

HTTPS/SSL Interception ☐ Enable SSL filtering.

SSL/MITM Mode Splice Whitelist, Bump Otherwise
 The SSL/MITM mode determines how SSL interception is treated when 'SSL Man In The Middle Filtering' is enabled.
 Default: Splice Whitelist, Bump Otherwise. [Click Info for details.](#)

SSL Intercept Interface(s) WAN
 LAN
 DMZ
 MANAGEMENT
 The interface(s) the proxy server will intercept SSL requests on. Use CTRL + click to select multiple interfaces.

SSL Proxy Port
 This is the port the proxy server will listen on to intercept SSL while using transparent proxy. Default: 3129

SSL Proxy Compatibility Mode Modern
 The compatibility mode determines which cipher suites and TLS versions are supported. Default: Modern. [Click Info for details.](#)

DHParams Key Size 2048 (default)
 DH parameters are used for temporary/ephemeral DH key exchanges and improve security by enabling the use of DHE ciphers.

CA none
 Select Certificate Authority to use when SSL interception is enabled.

SSL Certificate Daemon Children
 This is the number of SSL certificate daemon children to start. May need to be increased in busy environments. Default: 5

Remote Cert Checks Accept remote server certificate with errors
 Do not verify remote certificate
 Select remote SSL certificate checks to perform. Use CTRL + click to select multiple options.

Certificate Adapt Sets the "Not After" (setValidAfter)
 Sets the "Not Before" (setValidBefore)
 Sets CN property (setCommonName)
 See sslproxy_cert_adapt directive documentation and Mimic original SSL server certificate wiki article for details.

Logging Settings

Enable Access Logging ☐ This will enable the access log.
Warning: Do NOT enable if available disk space is low.

3.2. Onglet “Cache”

- Activer le cache disque

Package / Proxy Server: Cache Management / Local Cache

General Remote Cache **Local Cache** Antivirus ACLs Traffic Mgmt Authentication Users Real Time Status Sync

Squid Cache General Settings

Disable Caching ☐ Disable caching completely.
 This may be required if Squid is only used as a proxy to audit website access.

Cache Replacement Policy Heap LRU/LDA
 The cache replacement policy decides which objects will remain in cache and which objects are replaced to create space for the new objects. Default: heap LRU/LDA

Low-Water Mark in % 90
 The low-water mark for AUFS/UPS/diskd cache object eviction by the cache_replacement_policy algorithm.

High-Water Mark in % 95
 The high-water mark for AUFS/UPS/diskd cache object eviction by the cache_replacement_policy algorithm.

Do Not Cache
 Enter domain(s) and/or IP address(es) that should never be cached. Put each entry on a separate line.

Enable Offline Mode ☐ Enable this option and the proxy server will never try to validate cached objects.
 Offline mode gives access to more cached information than normally allowed (e.g., expired cached versions where the origin server should have been contacted otherwise).

External Cache Managers
 Enter the IPs for the external Cache Managers to be granted access to this proxy. Separate entries by semi-colons (,)

Squid Hard Disk Cache Settings

External Cache Managers

Enter the IPs for the external Cache Managers to be granted access to this proxy. Separate entries by semi-colons (,)

Squid Hard Disk Cache Settings

Hard Disk Cache Size
Amount of disk space (in megabytes) to use for cached objects.

Hard Disk Cache System
This specifies the kind of storage system to use. ⓘ

Clear Disk Cache NOW Hard Disk Cache is automatically managed by swapstate_check.php script which is scheduled to run daily via cron. ⓘ
If you wish to clear cache **immediately**, click this button **once**:

Level 1 Directories
Specifies the number of Level 1 directories for the hard disk cache. ⓘ

Hard Disk Cache Location
This is the directory where the cache will be stored. Default: /var/squid/cache ⓘ

Minimum Object Size
Objects smaller than the size specified (in kilobytes) will not be saved on disk. Default: 0 (meaning there is no minimum) ⓘ

Maximum Object Size
Objects larger than the size specified (in megabytes) will not be saved on disk. Default: 4 (MB) ⓘ

Squid Memory Cache Settings

Memory Cache Size
Specifies the ideal amount of physical RAM (in megabytes) to be used for In-Transit objects, Hot Objects and Negative-Cached objects.
Minimum value: 1 (MB). Default: 64 (MB) ⓘ

Maximum Object Size in RAM
Objects greater than this size (in kilobytes) will not be attempted to kept in the memory cache. Default: 256 (KB) ⓘ

Memory Replacement

Activer Windows
Accédez aux paramètres pour activer Windows.

4. Installation des listes SquidGuard

Menu : **Services** → **SquidGuard**

pfSense
COMMUNITY EDITION

System Interfaces Firewall Services VPN Status Diagnostics Help

Package / Proxy filter SquidGuard: General settings / General settings

General settings Common ACL Groups ACL Target categories Times Rewrites Blacklist Log XMLRPC Sync

General Options

Enable

☒ Check this option to enable squidGuard.
Important: Please set up at least one category on the Target Categories tab before enabling. See this link for details.
The Save button at the bottom of this page must be clicked to save configuration changes.
To activate squidGuard configuration changes, the Apply button must be clicked.

Apply

SquidGuard service state: **STARTED**

LDAP Options

Enable LDAP Filter

☐ Enable options for setup ldap connection to create filters with ldap search

LDAP DN

Configure your LDAP DN (ex: cn=Administrator,cn=Users,dc=domain)

LDAP DN Password

Password must be initialize with letters (Ex: Change123), valid format: [a-zA-Z][a-zA-Z0-9/_\.\!\@\#\%\^&]*

LDAP Cache Time

Number of seconds to cache LDAP Results (recommended value: 300)

Strip NT domain name

☐ Strip NT domain name component from user names (/ or \ separated).

Strip Kerberos Realm

☐ Strip Kerberos Realm component from user names (@ separated).

LDAP Version

Version 3

Service options

Activer Windows
Accédez aux paramètres pour activer Windows.

LDAP Version

Version 3

Service options

Rewrite process children

Maximum number of SquidGuard redirector processes that Squid may spawn. Using too few of these helper processes (a.k.a. "helpers") creates request queues. Using too many helpers wastes your system resources. (Default: 16)

Rewrite process children startup

Sets a minimum of how many SquidGuard processes are to be spawned when Squid starts or reconfigures. (Default: 8)

Rewrite process children idle

Sets a minimum of how many SquidGuard processes Squid is to try and keep available at all times. (Default: 4)

Logging options

Enable GUI log

☒ Check this option to log the access to the Proxy Filter GUI.

Enable log

☒ Check this option to log the proxy filter settings like blocked websites in Common ACL, Group ACL and Target Categories. This option is usually used to check the filter settings.

Enable log rotation

☐ Check this option to rotate the logs every day. This is recommended if you enable any kind of logging to limit file size and do not run out of disk space.

Miscellaneous

Clean Advertising

☐ Check this option to display a blank gif image instead of the default block page. With this option the user gets a cleaner webpage.

Blacklist options

Blacklist

☒ Check this option to enable blacklist

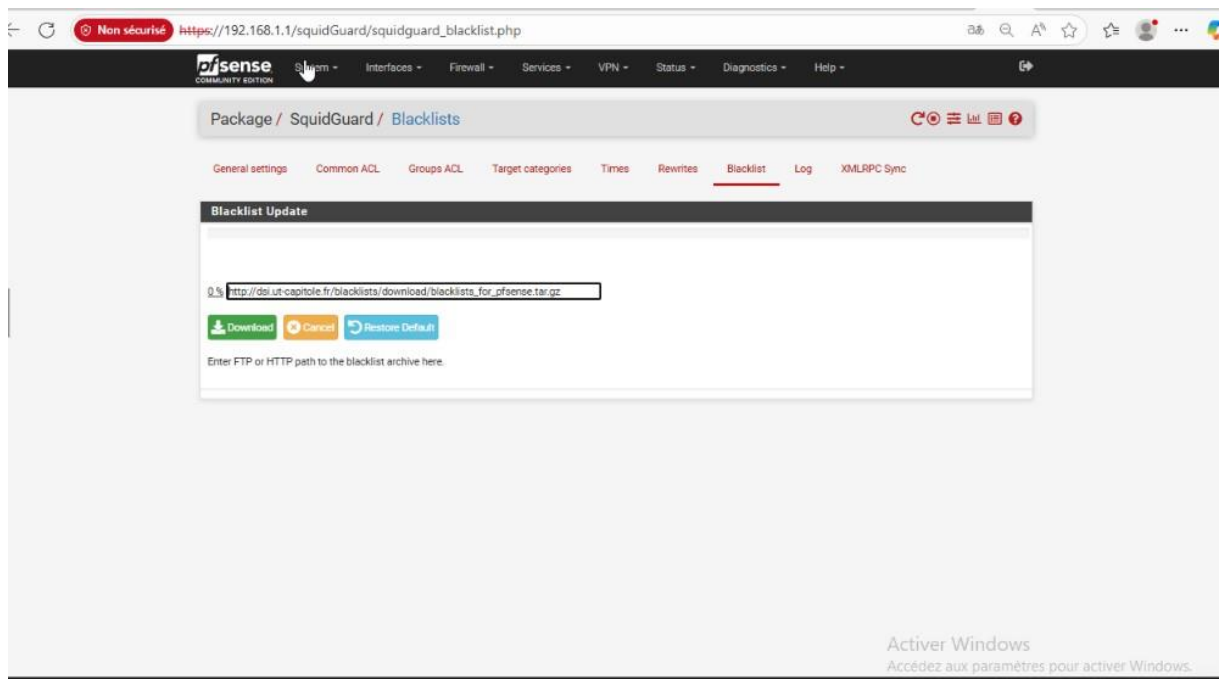
Blacklist proxy

Blacklist upload proxy - enter here, or leave blank.
Format: host[:port login:pass] - Default proxy port 1080.
Example: "192.168.0.1:8080 user:pass"

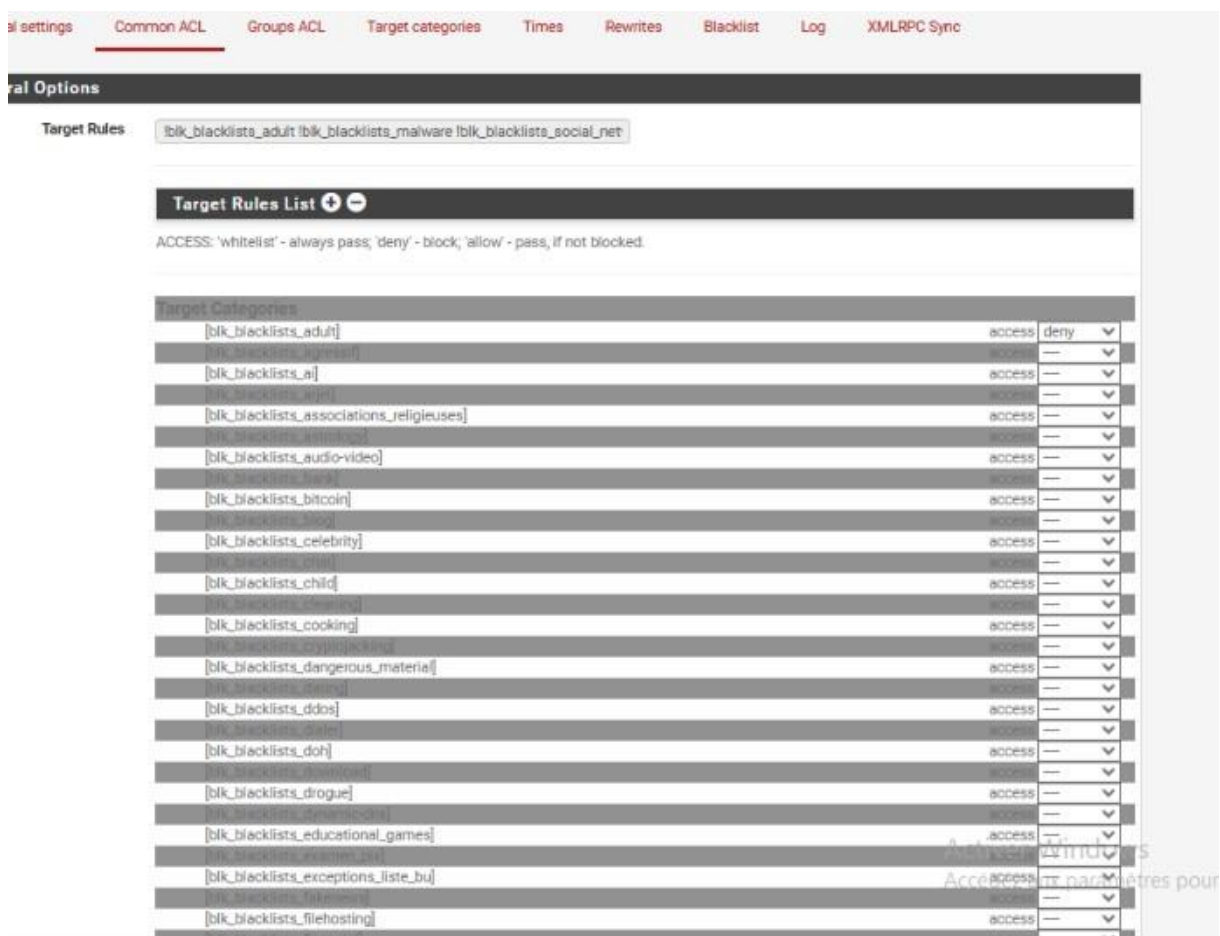
Blacklist URL

Enter the path to the blacklist (blacklist.tar.gz) here. You can use FTP HTTP or LOCAL URL blacklist archive or leave blank. The LOCAL path could be

Activer Windows
Accédez aux paramètres pour activer Windows.



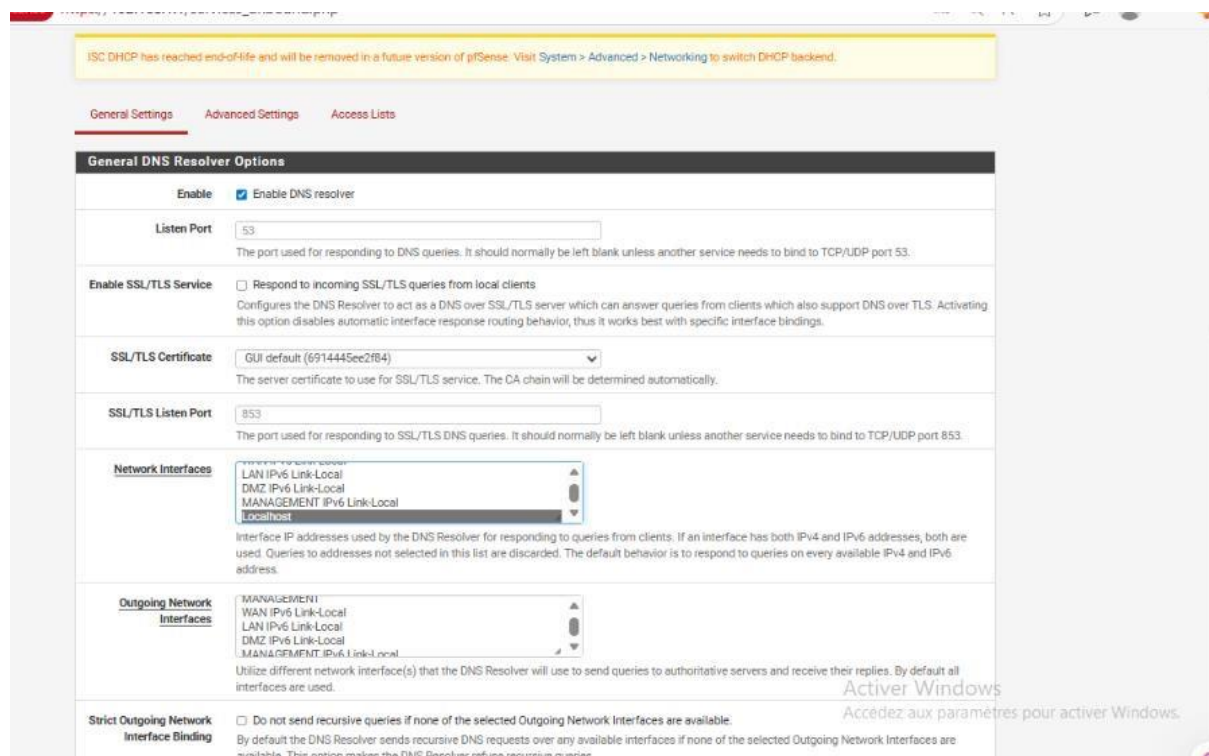
J'ai ensuite bloquer les 3 categorie de site qui ne doivent pas etre accessible



J'ai ensuite configuré la page du dns resolver, network interfaces j'ai choisi lan, wan et localhost, activer le dns resolver et activer le mode transparent

Dans la section host Overrides j'ai ajouter — Host : www — Domain : facebook.com — Ip Address : 127.0.0.1 comme l'objectif est de bloquer le site avant la connexion tls comme squidguard agit au niveau du proxy http mais Squid ne voit pas le nom du site, car le contenu est chiffré

Ensuite j'ai bien bloqué l'accès à facebook, j'ai testé sur une vm connecter à vmbr1, en ayant vidé le cache dns



Non sécuriséhttps://192.168.1.1/services_unbound.php

MANAGEMENT
WAN (IPv6 Link-Local)
LAN (IPv6 Link-Local)
DMZ (IPv6 Link-Local)
MANAGEMENT (IPv4 Link-Local)

Utilize different network interface(s) that the DNS Resolver will use to send queries to authoritative servers and receive their replies. By default all interfaces are used.

Strict Outgoing Network Interface Binding

☐ Do not send recursive queries if none of the selected Outgoing Network Interfaces are available.
By default, the DNS Resolver sends recursive DNS requests over any available interfaces if none of the selected Outgoing Network Interfaces are available. This option makes the DNS Resolver refuse recursive queries.

System Domain Local Zone Type

Transparent
The local zone type used for the pfSense system domain (System | General Setup | Domain). Transparent is the default.

DNSSEC

☒ Enable DNSSEC Support

Python Module

☐ Enable Python Module
Enable the Python Module.

DNS Query Forwarding

☐ Enable Forwarding Mode
If this option is set, DNS queries will be forwarded to the upstream DNS servers defined under System > General Setup or those obtained via dynamic interfaces such as DHCP, PPP, or OpenVPN (if DNS Server Override is enabled there).
☐ Use SSL/TLS for outgoing DNS Queries to Forwarding Servers
When set in conjunction with DNS Query Forwarding, queries to all upstream forwarding DNS servers will be sent using SSL/TLS on the default port of 853. Note that ALL configured forwarding servers MUST support SSL/TLS queries on port 853.

DHCP Registration

☒ Register DHCP leases in the DNS Resolver
If this option is set, then machines that specify their hostname when requesting an IPv4 DHCP lease will be registered in the DNS Resolver so that their name can be resolved. Note that this will cause the Resolver to reload and flush its resolution cache whenever a DHCP lease is issued. The domain in System > General Setup should also be set to the proper value.

Static DHCP

☒ Register DHCP static mappings in the DNS Resolver
If this option is set, then DHCP static mappings will be registered in the DNS Resolver, so that their name can be resolved. The domain in System > General Setup should also be set to the proper value.

OpenVPN Clients

☐ Register connected OpenVPN clients in the DNS Resolver
If this option is set, then the common name (CN) of connected OpenVPN clients will be registered in the DNS Resolver, so that their name can be resolved. This only works for OpenVPN servers (Remote Access SSL/TLS or User Auth with Username as Common Name option) operating in "tun" mode. The domain in System > General Setup should also be set to the proper value.

OpenVPN Clients

☐ Register connected OpenVPN clients in the DNS Resolver
If this option is set, then the common name (CN) of connected OpenVPN clients will be registered in the DNS Resolver, so that their name can be resolved. This only works for OpenVPN servers (Remote Access SSL/TLS or User Auth with Username as Common Name option) operating in "tun" mode. The domain in System > General Setup should also be set to the proper value.

Display Custom Options

☒ Display Custom Options

Save

Host Overrides

Host	Parent domain of host	IP to return for host	Description	Actions
pfsense	local	192.168.99.1	interface management	
webdmz	local	192.168.2.10		
www	facebook.com	127.0.0.1		

Enter any individual hosts for which the resolver's standard DNS lookup process should be overridden and a specific IPv4 or IPv6 address should automatically be returned by the resolver. Standard and also non-standard names and parent domains can be entered, such as 'test', 'has.home.arpa', 'mycompany.localdomain', '1.168.192.in-addr.arpa', or 'somesite.com'. Any lookup attempt for the host will automatically return the given IP address, and the usual lookup server for the domain will not be queried for the host's records.

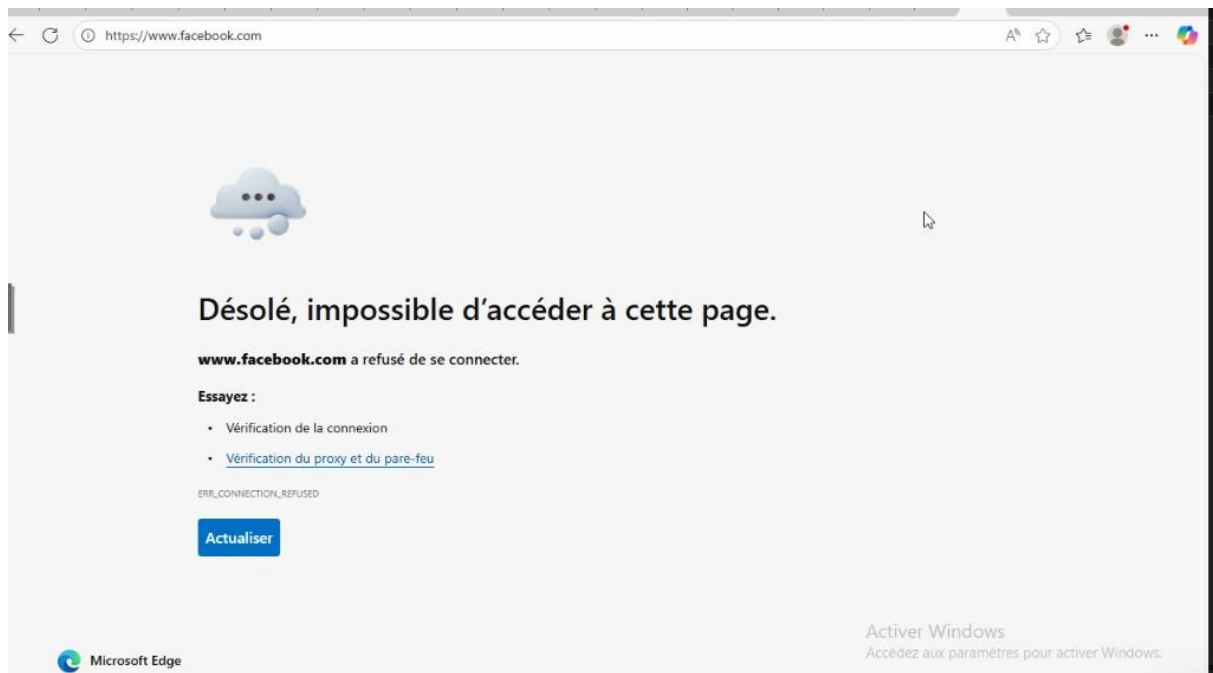
Add

Domain Overrides

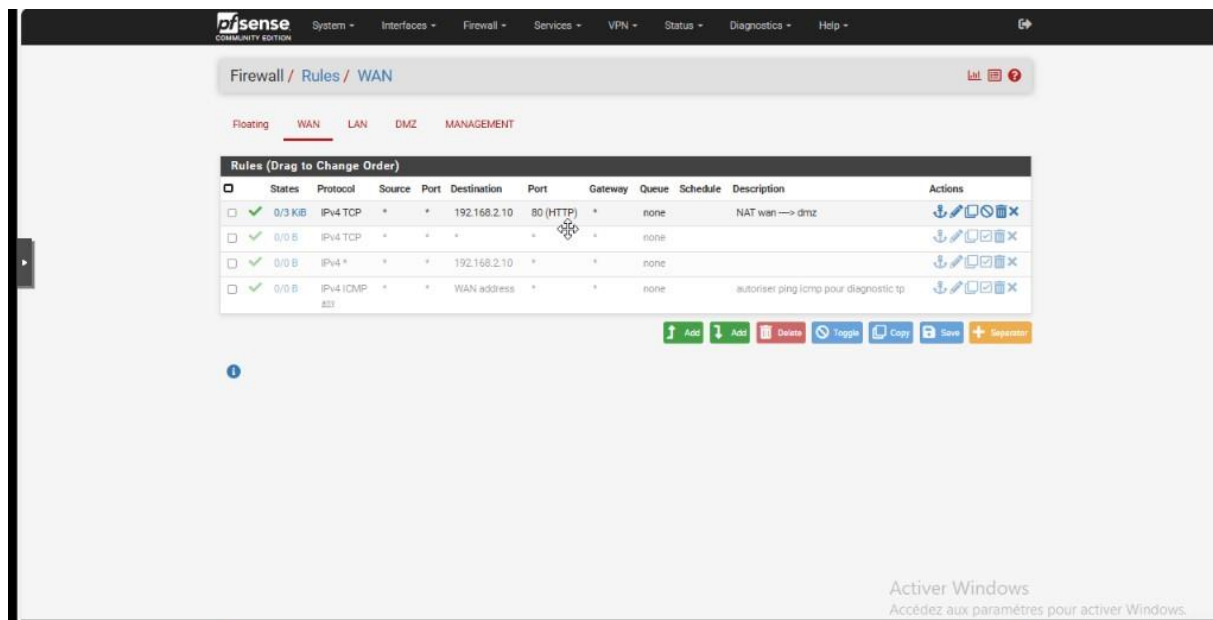
Domain	Lookup Server IP Address	Description	Actions
--------	--------------------------	-------------	---------

Enter any domains for which the resolver's standard DNS lookup process should be overridden and a different (non-standard) lookup server should be queried instead. Non-standard, 'invalid' and local domains, and subdomains, can also be entered, such as 'test', 'has.home.arpa', 'mycompany.localdomain', '1.168.192.in-addr.arpa', or 'somesite.com'. The IP address is treated as the authoritative lookup server for the domain (including all of its subdomains), and other lookup servers will not be queried. If there are multiple authoritative DNS servers available for a domain then make a separate entry for each, using the same domain name.

Add



Voici mes regles par feu :



pfSense COMMUNITY EDITION System Interfaces Firewall Services VPN Status Diagnostics Help

Firewall / Rules / LAN

Floating WAN LAN DMZ MANAGEMENT

Rules (Drag to Change Order)

☐	Status	Protocol	Source	Port	Destination	Port	Gateway	Queue	Schedule	Description	Actions
☐	✓ 27/70.47 MB	IPv4 *	LAN subnets	*	*	*	*	none		Default allow LAN to any rule	↓ ↗ ↘ ↻ ⌂
☐	✓ 0/0 B	IPv4 TCP	LAN subnets	*	192.168.2.10	web_ports	*	none		autoriser lan vers web dmz	↓ ↗ ↘ ↻ ⌂
☐	✗ 0/0 B	IPv4 TCP	LAN subnets	*	This Firewall (self)	80-443	*	none			↓ ↗ ↘ ↻ ⌂
☐	✗ 0/0 B	IPv4 *	192.168.1.1/24	*	192.168.2.0/24	*	*	none		blocker lan dmz	↓ ↗ ↘ ↻ ⌂
☐	✓ 0/0 B	IPv4 *	*	*	*	*	*	none			↓ ↗ ↘ ↻ ⌂
☐	✓ 0/0 B	IPv6 *	LAN subnets	*	*	*	*	none		Default allow LAN IPv6 to any rule	↓ ↗ ↘ ↻ ⌂
☐	✓ 0/0 B	IPv4 *	LAN subnets	*	*	*	*	none		autoriser tout vers Internet	↓ ↗ ↘ ↻ ⌂

↑ Add ↓ Add Delete Toggle Copy Save + Separator

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pfSense COMMUNITY EDITION System Interfaces Firewall Services VPN Status Diagnostics Help

Firewall / Rules / DMZ

Floating WAN LAN DMZ MANAGEMENT

Rules (Drag to Change Order)

☐	Status	Protocol	Source	Port	Destination	Port	Gateway	Queue	Schedule	Description	Actions
☐	✗ 0/2 KIB	IPv4 *	DMZ subnets	*	LAN subnets	*	*	none		blocage dmz → lan	↓ ↗ ↘ ↻ ⌂
☐	✓ 0/3 KIB	IPv4 *	DMZ subnets	*	*	*	*	none		autoriser dmz vers Internet	↓ ↗ ↘ ↻ ⌂
☐	✗ 0/0 B	IPv4 *	DMZ subnets	*	LAN subnets	*	*	none		isoler la dmz du lan	↓ ↗ ↘ ↻ ⌂
☐	✗ 0/0 B	IPv4 TCP	DMZ subnets	*	LAN subnets	*	*	none			↓ ↗ ↘ ↻ ⌂
☐	✓ 0/0 B	IPv4 *	*	*	*	*	*	none		autoriser sortie dmz pour serveur debian	↓ ↗ ↘ ↻ ⌂
☐	✓ 0/0 B	IPv4 *	DMZ subnets	*	DMZ address	*	*	none			↓ ↗ ↘ ↻ ⌂
☐	✓ 0/0 B	IPv4 TCP	192.168.2.10/24	*	DMZ address	*	*	none			↓ ↗ ↘ ↻ ⌂
☐	✓ 0/0 B	IPv4 TCP	DMZ address	*	192.168.2.10/24	*	*	none			↓ ↗ ↘ ↻ ⌂

↑ Add ↓ Add Delete Toggle Copy Save + Separator

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Firewall / Rules / MANAGEMENT

Floating WAN LAN DMZ MANAGEMENT

Rules (Drag to Change Order)

☐	Status	Protocol	Source	Port	Destination	Port	Gateway	Queue	Schedule	Description	Actions
☐	✓ 0/0 B	IPv4 TCP	MANAGEMENT subnets	*	This Firewall (self)	*	*	none			↓ ↗ ↘ ↻ ⌂
☐	✓ 0/463.95 MB	IPv4 *	192.168.99.0/24	*	*	*	*	none			↓ ↗ ↘ ↻ ⌂

↑ Add ↓ Add Delete Toggle Copy Save + Separator

IV.2. Installation du serveur de logs (Rsyslog + WebUI)

1 — Installer un serveur Debian “LogServer”

J'ai créer une VM Debian

Connectée au **vmbr1 (LAN)**

, j'ai ensuite configurer l'ip, dns et gateway etc, en allant sur le sudo

nano /etc/network/interfaces Et renseigner : auto ens18 iface

ens18 inet static address 192.168.1.50 netmask 255.255.255.0

gateway 192.168.1.1 dns-nameservers 1.1.1.1



```
root@debian2:~# ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen
link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
inet 127.0.0.1/8 scope host lo
    valid_lft forever preferred_lft forever
inet6 ::1/128 scope host noprefixroute
    valid_lft forever preferred_lft forever
2: ens18: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group def
link/ether bc:24:11:58:78:54 brd ff:ff:ff:ff:ff:ff
altname enp0s18
inet 192.168.1.50/24 brd 192.168.1.255 scope global ens18
    valid_lft forever preferred_lft forever
root@debian2:~#
```

J'ai ensuite Installer et configurer Rsyslog avec les commande sudo apt update

sudo apt install rsyslog -y, j'ai editer le fichier de configuration nano avec sudo nano /etc/rsyslog.conf

j'ai decommenté provides udp syslog reception et provides tcp syslog reception


```
Non sécurisé https://100.125.89.82:8006/?console=kvm&novnc=1&vmid=102
GNU nano 7.2 /etc/rsyslog
# /etc/rsyslog.conf configuration file for rsyslog
#
# For more information install rsyslog-doc and see
# /usr/share/doc/rsyslog-doc/html/configuration/index.html
#####
#### MODULES ####
#####

module(load="imuxsock") # provides support for local system logging
module(load="imklog")   # provides kernel logging support
#module(load="immark")  # provides --MARK-- message capability

# provides UDP syslog reception
#module(load="imudp")
#input(type="imudp" port="514")

# provides TCP syslog reception
module(load="imtcp")
input(type="imtcp" port="514")

#####
#### GLOBAL DIRECTIVES ####
#####

#
# Set the default permissions for all log files.
#
$FileOwner root
$FileGroup adm
$FileCreateMode 0640
$DirCreateMode 0755
$Umask 0022

#
# Where to place spool and state files
#
$WorkDirectory /var/spool/rsyslog

#
# Include all config files in /etc/rsyslog.d/
#
$IncludeConfig /etc/rsyslog.d/*.conf
```

J'ai ensuite redémarré le service : `sudo systemctl restart rsyslog`

Et enfin j'ai vérifié que le port écoute avec la commande `sudo netstat -tulpn | grep 514`, yes les ports 514 sont ouverts

```

input(type="imudp" port="514")
    provides TCP syslog reception
module(load="imtcp")
re input(type="imtcp" port="514")

38 #####
#### GLOBAL DIRECTIVES ####
#####

#
# Set the default permissions for all log files.
#
$FileOwner root
$FileGroup adm
$FileCreateMode 0640
$DirCreateMode 0755
$Umask 0022

# Where to place spool and state files
$WorkDirectory /var/spool/rsyslog

X#
# Include all config files in /etc/rsyslog.d/
#
$IncludeConfig /etc/rsyslog.d/*.conf

X#####
#### RULES ####
#####

#
X# Log anything besides private authentication messages to a single log file
#
*.:*;auth,authpriv.none -/var/log/syslog

root@debian2:~# sudo ss -tulpn | grep 514
Xroot@debian2:~# sudo ss -tulpn | grep 514
root@debian2:~# sudo systemctl restart rsyslog
root@debian2:~# sudo ss -tulpn | grep 514
)udp UNCONN 0 0 0.0.0.0:514 0.0.0.0:* users:(("rsyslogd",pid=1452,fd
udp UNCONN 0 0 [::]:514 [::]:* users:(("rsyslogd",pid=1452,fd
tcp LISTEN 0 25 0.0.0.0:514 0.0.0.0:* users:(("rsyslogd",pid=1452,fd
tcp LISTEN 0 25 [::]:514 [::]:* users:(("rsyslogd",pid=1452,fd
root@debian2:~#

```

IV.3. Configuration pfSense → Serveur de logs